Polygenic Risk Score Portability on Real Data: Alzheimer's Disease

BIOT 6900 · Module 6 Assignment · Bhuvanesh Karthik Nilavalagan

Background

This module involves applying an existing Alzheimer's PGS (PGS000334 in the PGS Catalog, with 22 SNPs, including both APOE variants that define ε2 and ε4) to real genotypes from 1000 Genomes Project in 2,336 samples of five different ancestries. It seems to be straightforward: a PGS based on the majority of European-ancestry studies, would it perform equally well for all of us?

First of all, it has to be mentioned that the genotypes, the weightings of the SNPs, and the ancestries are all real, however, the disease status (whether the person is the case or the control) is simulated, since 1000 Genomes does not provide any disease information for its participants. So, the distribution of the scores among the groups is a real result, however, the values of AUC and calibration are illustrative, since they depend on the simulated label rather than real diagnoses.

Proof of real data (Section 2 of the notebook):

source : REAL
people : 2336
score size : 22 variants
by ancestry : {'AFR': 500, 'EAS': 500, 'EUR': 500, 'SAS': 489, 'AMR': 347}
genotypes.csv: sha256 9cf41a1f
variants.csv : sha256 f7a87d3f
score.csv : sha256 7f80e8ff
meta.csv : sha256 6cc55b6e

Method

The arithmetic of a PRS is just dosages times weights, summed. Everything difficult lives in the bookkeeping. The first step was allele harmonization: the score lists an “effect allele” for each variant, but the genotype file counts whichever allele 1000 Genomes calls ALT. If those two don't match, the variant counts the wrong allele and its effect goes in the wrong direction. For each variant I kept it as is if the effect allele was ALT, flipped the dosage (2 − dosage) if it was REF, and dropped it if it matched neither.

Result: all 22 variants used, 13 flipped, 0 dropped. Thirteen of 22 is more than half, which surprised me. Without this step, more than half of the score would have been pointing the wrong way, and the PRS would have been quietly wrong without any error message.

All 22 of the score’s variants were also present in the genotype panel, so none were silently excluded for being absent, which matters because a variant dropped for allele mismatch and a variant missing from the data are two different failures and only the first one produces a warning.

I then computed the PRS for everyone, standardized it once across the whole cohort (not separately per group, which would hide the differences), and ran four checks: the PRS distribution per ancestry, what one fixed top-10% cutoff does to each group, the AUC within each group, and calibration of a risk model fitted only in Europeans.

1. Portability: why the score loses accuracy outside Europeans

	EUR	AFR	EAS	SAS	AMR	Overall
AUC	0.672	0.493	0.575	0.611	0.555	0.599

The score works best in Europeans (AUC 0.67) and worse everywhere else, down to 0.49 in African-ancestry samples, which is no better than a coin flip. I expected EUR to come out on top, but the drop in AFR was bigger than I predicted.

This is because most of the variants in the PRS are not the causative variants themselves. Rather, these are "tags" variants that lie near the causative variants on the chromosome and have a high probability of being passed down alongside the latter. Linkage Disequilibrium (LD), how closely a tag follows the causative variant differs between populations. The PRS itself was developed in Europeans, thus, the tag selection is based on the tags that best follow causative variants among Europeans. In the simulated data, the mean tag-causative correlation (r^2) in EUR was 0.82; in EAS, SAS and AMR it was about 0.73-0.75; and in AFR, it was 0.44. The genomes in Africa ancestry populations are the oldest and most genetically diverse, thus, LD blocks are small, and it is likely that the European tag would drift away from its causative partner. This ranking perfectly matches the AUC ranking.

Allele frequencies also differ between groups, which shifts the whole score up or down. The mean PRS was +0.74 in AFR but −0.47 in EAS, even though nothing about disease risk was used to calculate those numbers.

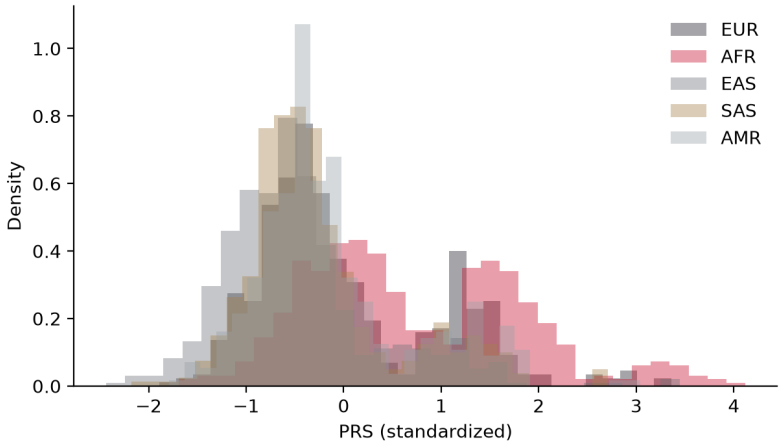


Figure 1. Standardized PRS by ancestry (real genotypes and weights, no phenotype used).

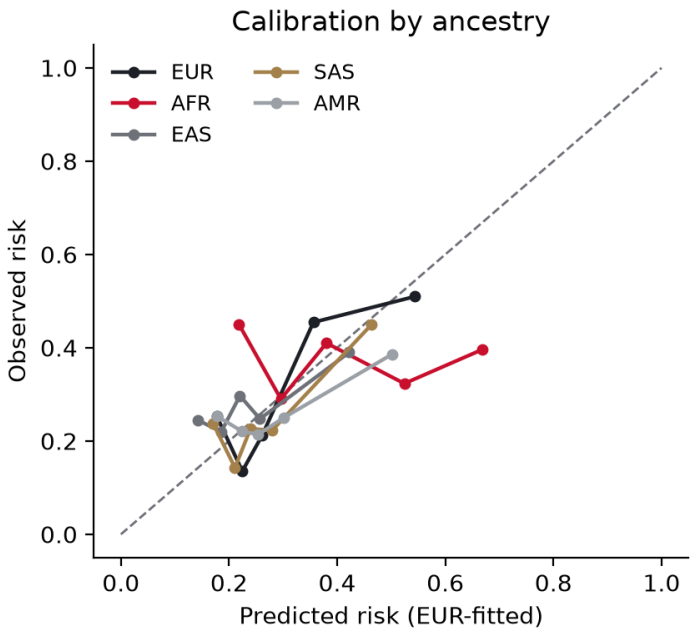


Figure 2. Calibration of a model fitted in EUR only, applied to every group.

2. One threshold, many groups

	EUR	AFR	EAS	SAS	AMR
Mean PRS	+0.04	+0.74	−0.47	−0.25	−0.10
% above pooled top-10% cutoff	7.8	28.0	2.2	4.3	6.3

No, a single top-10% cutoff is not fair. If a clinic took the score's top 10% as “high risk” and applied it unchanged, 28% of African-ancestry patients would be flagged, compared with 2.2% of East Asian patients, a thirteen-fold difference. The worst part is that the group flagged most often (AFR) is exactly the group where the score has no ability to tell cases from controls (AUC 0.49). So those 28% would be labelled high risk mostly because of allele-frequency differences, not because they are actually more likely to develop the disease. Meanwhile, a genuinely high-risk East Asian patient could easily fall below the cutoff and be missed.

This problem appears before any disease label is involved. Figure 1 uses only real genotypes and real weights, so the unfairness of the cutoff is a real finding, not an artifact of the simulated phenotype.

3. Discrimination vs. calibration

	EUR	AFR	EAS	SAS	AMR
Mean predicted risk	0.314	0.418	0.246	0.273	0.293
Observed case rate	0.314	0.374	0.280	0.256	0.265

The two checks answer different questions. Discrimination (AUC) asks: within a group, does the score rank cases above controls? Calibration asks: when the model says “30% risk”, do about 30% of those people actually become cases?

In EUR, where the model was fitted, the average predicted risk matched the observed rate exactly (0.314 vs. 0.314). Everywhere else it drifted. It over-predicted risk in AFR (0.42 vs. 0.37) and AMR, and under-predicted it in EAS (0.25 vs. 0.28). The AFR curve in Figure 2 is also nearly flat: its highest-predicted group (about 67% predicted) had an observed rate of only about 40%, which is lower than its lowest-predicted group (about 45%). In plain terms, the numbers the model gives an African-ancestry patient are both too high and not ranked in a meaningful order.

Where they disagreed was the most interesting part. SAS had a fairly reasonable AUC (0.61) but was still slightly over-predicted, while EAS had a lower AUC (0.58) but its average risk was off in the opposite direction. So a score can rank people reasonably well and still give them the wrong absolute risk, or vice versa. Checking only the AUC, which is what most papers report, would have missed the calibration problem completely. Calibration could only be tested this way because the model was fitted in one group and tested in the others; fitting and testing on the pooled data would have averaged the problem away.

4. Clinical utility and limits

To test whether the PRS provides additional information over a baseline clinical covariate, I compared an age-only model to an age-plus-PRS model (using 5-fold cross-validation). The area under the curve (AUC) increased from 0.562 to 0.618. This reflects a genuine but small gain that can be reasonably expected of a common-disease PRS, which provides only part of the information needed to make a decision.

Where it can be useful: In research environments, such as the enrichment of European-ancestry patients with increased genetic susceptibility for preventive trials, or correction for genetic background in studies. Because PRS provide their utility on the tails of the distribution, they serve better for enrichment rather than prediction.

Where it should never be applied: As a diagnostic test; as a cutoff that applies universally across ancestry groups; or for any decision made in the context of a patient who is not of European heritage without validation in that patient population. In the specific case of AD, there is currently also no treatment that would be indicated by a high score.

What I would need before deploying it: real diagnosed Alzheimer's cases and controls, not simulated labels, from each ancestry group the score would be used in. Examples include the Alzheimer's Disease Sequencing Project (ADSP), the All of Us Research Program, and the Million Veteran Program, which all include substantial non-European participants. With that, I would check AUC

and calibration within each group, recalibrate the model per ancestry, and ideally re-estimate the weights with multi-ancestry GWAS rather than relying on European weights alone.

5. Artifact vs. biology: APOE ε4 in the African-ancestry group

The score ran at chance in AFR here, partly because of how the phenotype was simulated. Each score variant was given a hidden "causal" partner chosen for being linked to it in Europeans. For APOE ε4, the causal partner chosen this way is almost completely unlinked to ε4 in African-ancestry samples ($r^2 \approx 0.01$). So in this simulated world, carrying ε4 barely changes an African-ancestry person's simulated disease risk, and the score loses its single most important variant in that group.

That is a property of the simulation design, not of real biology. In real studies, APOE ε4 is a risk allele in African-ancestry people too, although its effect is weaker on average than in Europeans (Farrer et al., 1997). The lesson I take from this is that a result can look like a biological finding ("ε4 doesn't matter in Africans") when it is really a statement about how the data were built. If I had not read the dataset's documentation, I could easily have drawn a false and harmful conclusion from a correct-looking number.

6. Ethics

However, this issue is more than just a technical challenge. The majority of genetic research has been performed on populations of European descent, thus creating prediction scores that work best for the populations already best served by medicine, and worst for those who are not (Martin et al., 2019). If such scores are implemented clinically as is, this could serve to exacerbate existing inequalities in health care rather than reduce them, through misidentifying African-ancestry patients for follow-up and worry while neglecting potential problems with East Asian patients.

Individual-level considerations also exist. The Alzheimer's risk score, particularly when the status of APOE is included in the risk score, is considered sensitive data, meaning that the data not only pertains to an individual's future but also to their relatives. Informed consent, genetic counseling, and proper communication that the risk score does not equate to diagnosis but indicates higher risk should take place. Lastly, since the phenotype in this database is generated, none of the actual data presented here can be cited as true Alzheimer's risk.

Summary

This harmonization phase was more important than I thought, with 13 of 22 variants needing to be reversed. The score developed by the European-trained model performed the best discrimination in EUR (AUC 0.67) and the worst in AFR (0.49), depending on how well the tagging variants were correlated with the causal variants within each population. Using one cut-off value would identify 28% of AFRs but 2.2% of EAS. Furthermore, the model calibrated in EUR would produce the wrong absolute risk estimates outside that population. When including the PRS with age in the prediction, there would be an actual improvement in performance (0.56 to 0.62).

References

1. Zhang, Q., et al. (2020). Risk prediction of late-onset Alzheimer's disease implies an oligogenic architecture. Nature Communications, 11, 4799. https://doi.org/10.1038/s41467-020-18534-1 (PGS Catalog PGS000334)

2. The 1000 Genomes Project Consortium (2015). A global reference for human genetic variation. Nature, 526, 68–74. https://doi.org/10.1038/nature15393

3. Martin, A. R., Kanai, M., Kamatani, Y., Okada, Y., Neale, B. M., & Daly, M. J. (2019). Clinical use of current polygenic risk scores may exacerbate health disparities. Nature Genetics, 51, 584–591. https://doi.org/10.1038/s41588-019-0379-x